edunet foundation

IBM SKILLSBUILD UNIVERSITY EDUCATION

Problem Statement No. 8 — Nutrition Agent

> Mandatory Technology: Use of IBM Cloud Lite services / IBM Granite is mandatory.

The Challenge

In an era where health awareness is growing, individuals increasingly seek personalized nutrition guidance. However, most existing tools provide generic diet plans, lack real-time adaptability, and fail to consider a person's holistic lifestyle, cultural preferences, allergies, and evolving health conditions. Furthermore, dieticians and nutritionists face limitations in scaling personalized consultations due to time and resource constraints.

Generative AI presents a groundbreaking opportunity to revolutionize this space by enabling an intelligent, interactive, and adaptive virtual nutrition assistant. By leveraging natural language processing (NLP), multimodal understanding, and large-scale dietary databases, an AI-powered assistant can generate dynamic meal plans, recommend smart food swaps, and explain nutritional choices—all tailored to the individual.

Project Objective

This project aims to develop **"The Smartest AI Nutrition Assistant"** using state-of-the-art generative AI models that:

- **Understand user inputs** via text, voice, or image (e.g., food photos, grocery labels).
- **Generate personalized meal plans** based on health goals, medical conditions, fitness routines, and preferences.
- **Offer contextual explanations** (e.g., "Why is this food better?").
- **Adapt suggestions dynamically** with continuous feedback.

By integrating health data, food databases, and LLM-powered reasoning, the solution will bridge the gap between one-size-fits-all diet apps and in-person nutrition counselling—delivering an AI that **thinks, learns, and cares like a real nutrition expert**.